

Cloud Computing in IoT

Subject – Internet of Things

CSE-5th SEM

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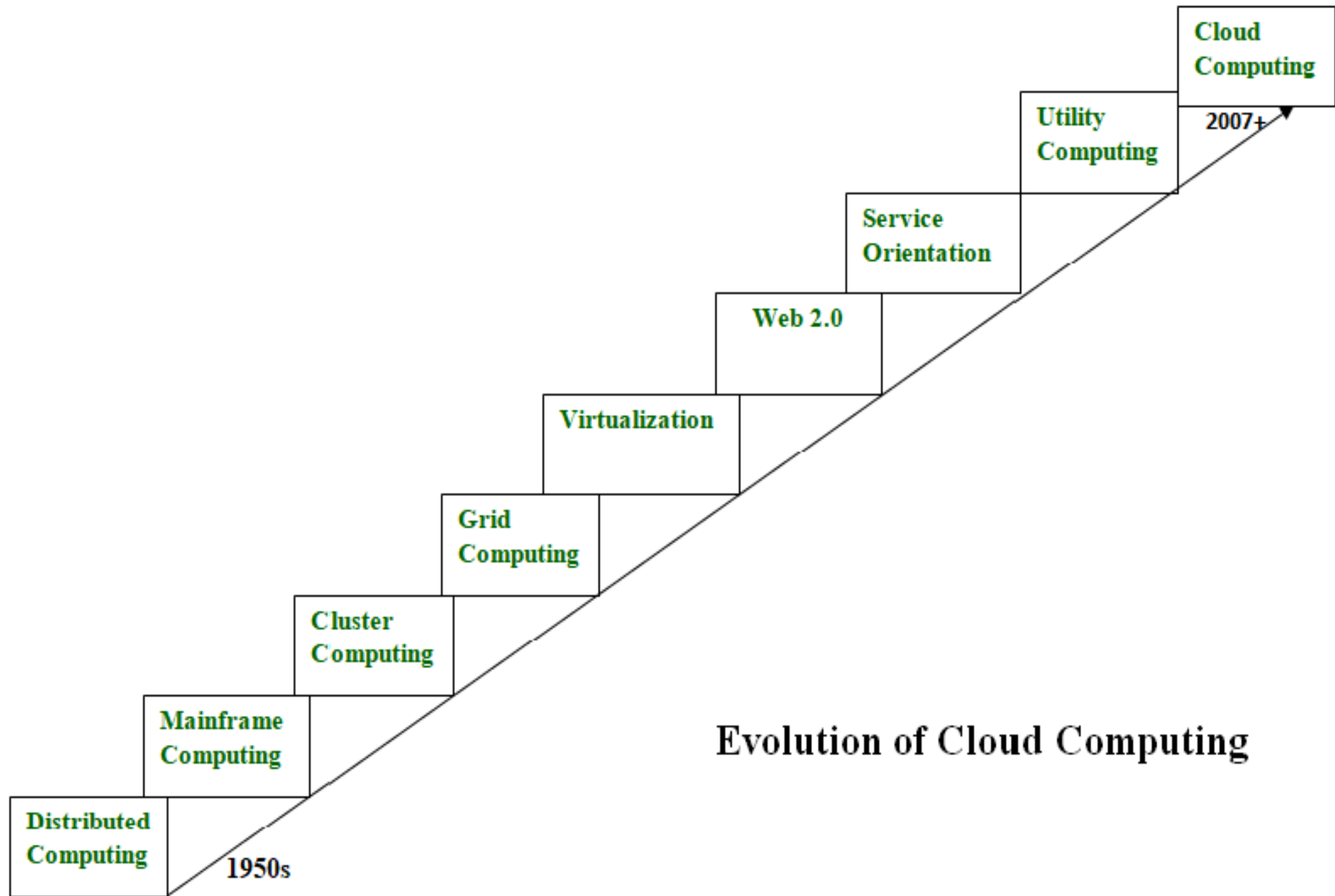
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Cloud Computing:

Cloud computing is the delivery of computing services— including

- servers
- storage
- databases
- networking
- software
- Analytics
- intelligence

over the Internet (“the cloud“) to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping lower your operating costs, run your infrastructure more efficiently and scale as your business needs change.



Evolution of Cloud Computing

	Traditional Computing	Cloud Computing
1	Data storage on local Internal / hard drive	Data storage in the cloud or external server
2	Extra storage – pay for hardware, no unlimited storage	Unlimited storage – no extra hardware required
3	For hardware configuration, you may hire expert	No need to do hardware configuration – dynamic resources
4	Pay for power, electricity, AC cooling etc.	No expense on power, electricity, AC cooling etc.
5	If you need to add new server it will take few days and more charges apply.	Pay for use, dynamic resources allocation.
6	Fixed charges – installation, hardware used.	If resources are use less, then pay less
7	Lack of isolation – high risk of security and poor performance.	No need to worry about security / loss of data.
8	Backup and recovery take risky.	Backup and recovery very easy.
9	Extra pay for software installation & upgrade.	Automatic software upgrade available
10	Pay for extra hardware & other resources	On demand resources

☐ **Cloud Computing architecture:**

Cloud Computing architecture comprises of many cloud components, which are loosely coupled. We can broadly divide the cloud architecture into two parts:

- ☐ Front End

- ☐ Back End

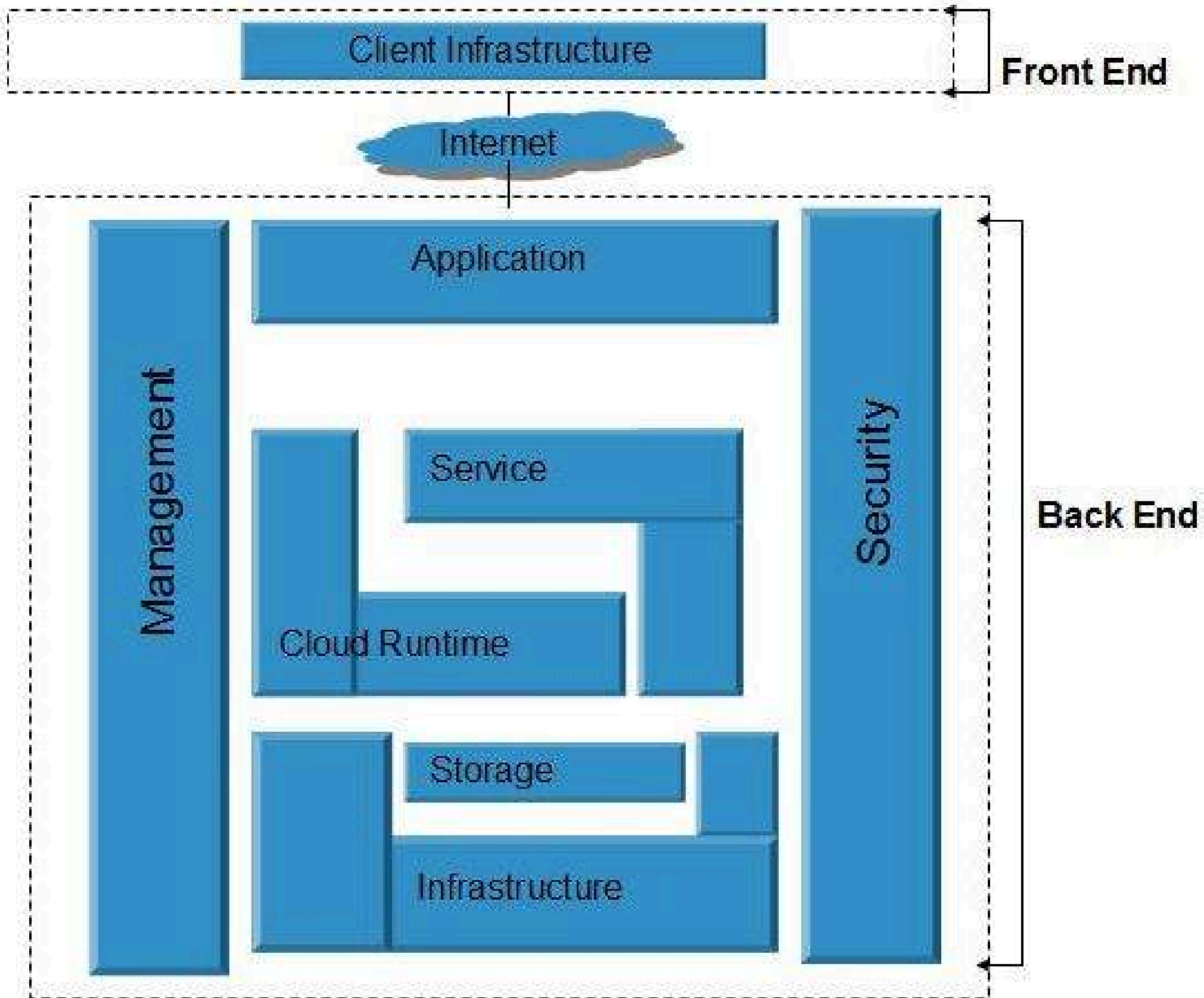
Each of the ends is connected through a network, usually Internet.

- ☐ **Front End**

The **front end** refers to the client part of cloud computing system. It consists of interfaces and applications that are required to access the cloud computing platforms, Example – Web Browser.

- ☐ **Back End**

The **back End** refers to the cloud itself. It consists of all the resources required to provide cloud computing services. It comprises of huge data storage, virtual machines, security mechanism, services, deployment models, servers, etc.



A list of advantages and disadvantages of cloud computing:

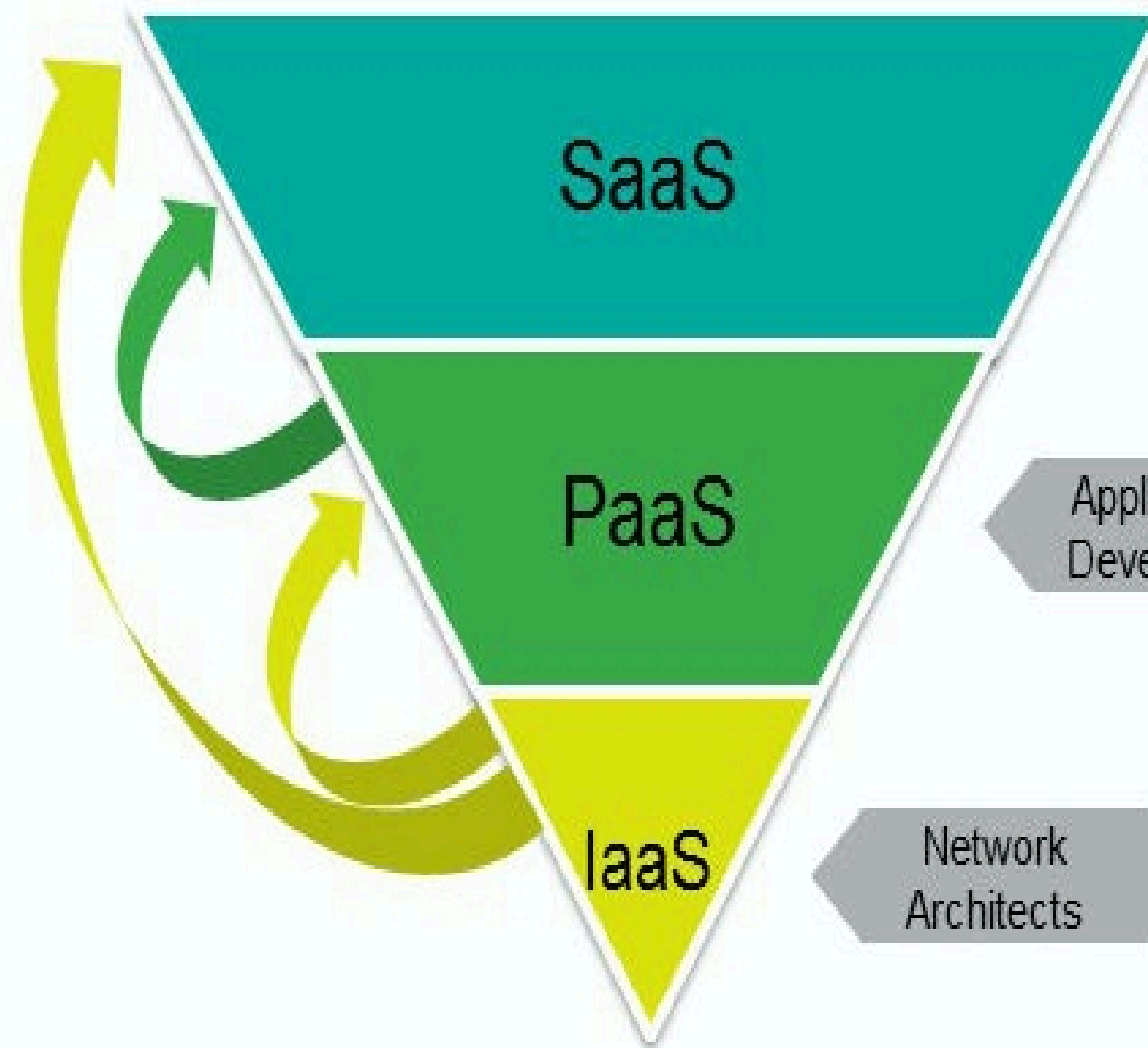
Advantages of Cloud	Disadvantages of Cloud
No cost of infrastructure	Good internet connection & bandwidth required
Minimum management and cost	Downtime
No administrative or management hassles	Limited control of infrastructure
Easy accessibility	Restricted or limited flexibility
Pay per use	Ongoing costs
Reliability	Security
Data control	Vendor lock-in
Data backup and recovery	Technical Issues
Huge cloud storage	NA



Service Models: The many paradigms of cloud computing can be broken down into three unique service model classifications:

- ☐ Software as a Service (SaaS)
- ☐ Platform as a Service (PaaS)
- ☐ Infrastructure as a Service (IaaS)

Value Visibility to End Users



End Users

Application
Developers

Network
Architects

□ What is Software-as-a-Service (SaaS)?

Software-as-a-Service, or SaaS for short, is a cloud-based method of providing software to users. SaaS users subscribe to an application rather than purchasing it once and installing it. Users can log into and use a SaaS application from any compatible device over the Internet. The actual application runs in cloud servers that may be far removed from a user's location.

Non-SaaS Application

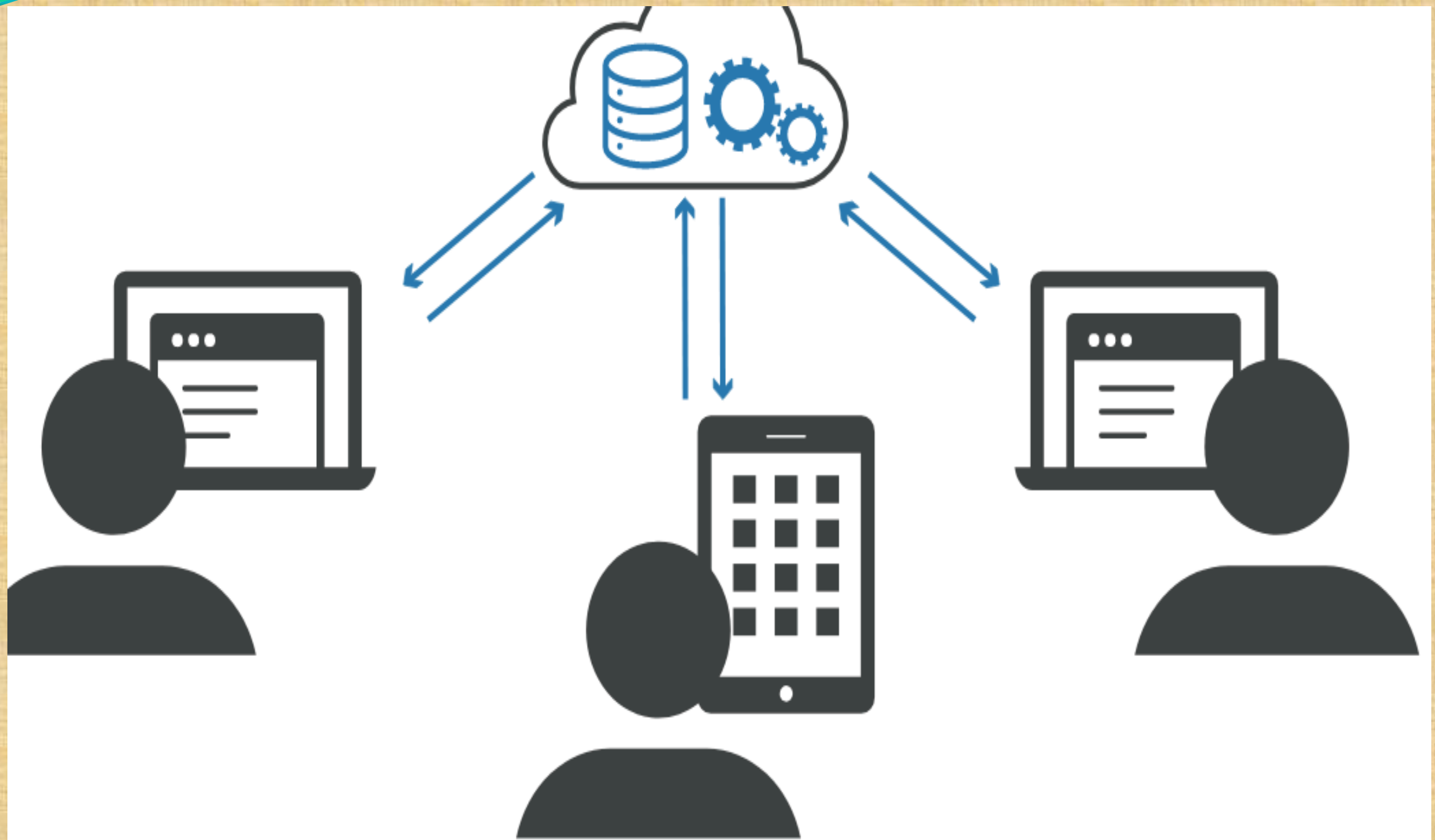


Application logic runs
on user's computer

SaaS Application



Application logic runs
in the cloud



Users can access SaaS applications on any device

- A SaaS application may be accessed through a browser or through an app. Online email applications that users access through a browser, such as Gmail and Office 365, are common examples of SaaS applications.
- The difference between SaaS and a software installation on a user's computer is somewhat like the difference between streaming a TV show online and buying all the seasons of the TV show on DVD.
- Someone who buys a TV show on DVD only needs to pay for it once; however, they will need to store and maintain the DVDs, and if they change their hardware – for instance, if they replace their DVD player with a Blu-ray player – then they will need to purchase the physical media again. Streaming the show instead means a third party handles all the storage and upgrades, and all a user needs to do is press play. However, streaming is dependent on an Internet connection, and users typically need to pay a recurring monthly fee to maintain their access.

What are the advantages and disadvantages of using SaaS?

- The SaaS model has a number of pros and cons, although for modern businesses and users the pros of SaaS often outweigh the cons. Here are some of the advantages and disadvantages of using SaaS applications:

Advantage:

- **Access from anywhere, on any device.** Typically, users can log into SaaS applications from any device and any location. This offers a great deal of flexibility – businesses can allow employees to operate all over the world, and users can access their files no matter they are. In addition, most users use multiple devices and replace them often; users don't need to reinstall SaaS applications or purchase new licenses each time they switch to a new device.
- **No need for updates or installations.** The SaaS provider updates and patches the application on an ongoing basis.
- **Scalability.** The SaaS provider handles scaling up the application, such as adding more database space or more compute power as usage increases.
- **Cost savings.** SaaS cuts down on internal IT costs and overhead. The SaaS provider maintains the servers and infrastructure that support the application, and the only cost to a business is the subscription cost of the application.

Infrastructure as a Service | IaaS

- IaaS is also known as **Hardware as a Service (HaaS)**. It is one of the layers of the cloud computing platform. It allows customers to outsource their IT infrastructures such as servers, networking, processing, storage, virtual machines, and other resources. Customers access these resources on the Internet using a pay-as-per use model.
- In traditional hosting services, IT infrastructure was rented out for a specific period of time, with pre-determined hardware configuration. The client paid for the configuration and time, regardless of the actual use. With the help of the IaaS cloud computing platform layer, clients can dynamically scale the configuration to meet changing requirements and are billed only for the services actually used.

A cloud computing service provider, such as Azure, manages the infrastructure, while you purchase, install, configure and manage your own software—operating systems, middleware and applications.

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- IaaS provider provides the following services –
 - **Compute:** Computing as a Service includes virtual central processing units and virtual main memory for the Vms that is provisioned to the end-users.
 - **Storage:** IaaS provider provides back-end storage for storing files.
 - **Network:** Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the Vms.
 - **Load balancers:** It provides load balancing capability at the infrastructure layer.

Advantages of IaaS cloud computing layer

There are the following advantages of IaaS computing layer –

☐ **1. Shared infrastructure**

IaaS allows multiple users to share the same physical infrastructure.

☐ **2. Web access to the resources**

IaaS allows IT users to access resources over the internet.

☐ **3. Pay-as-per-use model**

IaaS providers provide services based on the pay-as-per-use basis. The users are required to pay for what they have used.

☐ **4. Focus on the core business**

IaaS providers focus on the organization's core business rather than on IT infrastructure.

☐ **5. On-demand scalability**

On-demand scalability is one of the biggest advantages of IaaS. Using IaaS, users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

Disadvantages of IaaS:

☐ **1. Security**

Security is one of the biggest issues in IaaS. Most of the IaaS providers are not able to provide 100% security.

☐ **2. Maintenance & Upgrade**

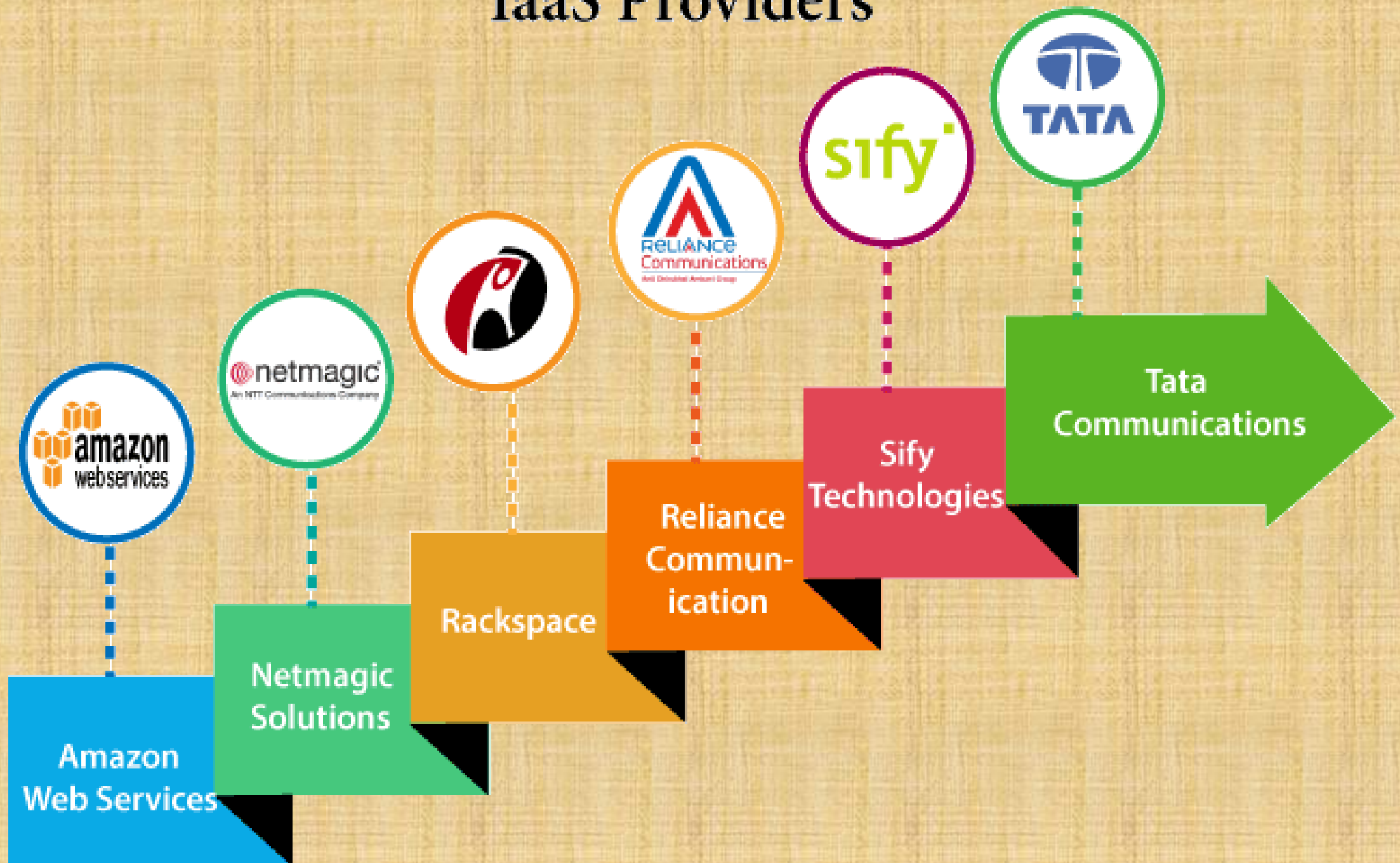
Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.

☐ **3. Interoperability issues**

It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in.

Top IaaS Providers who are providing IaaS cloud computing platform

IaaS Providers



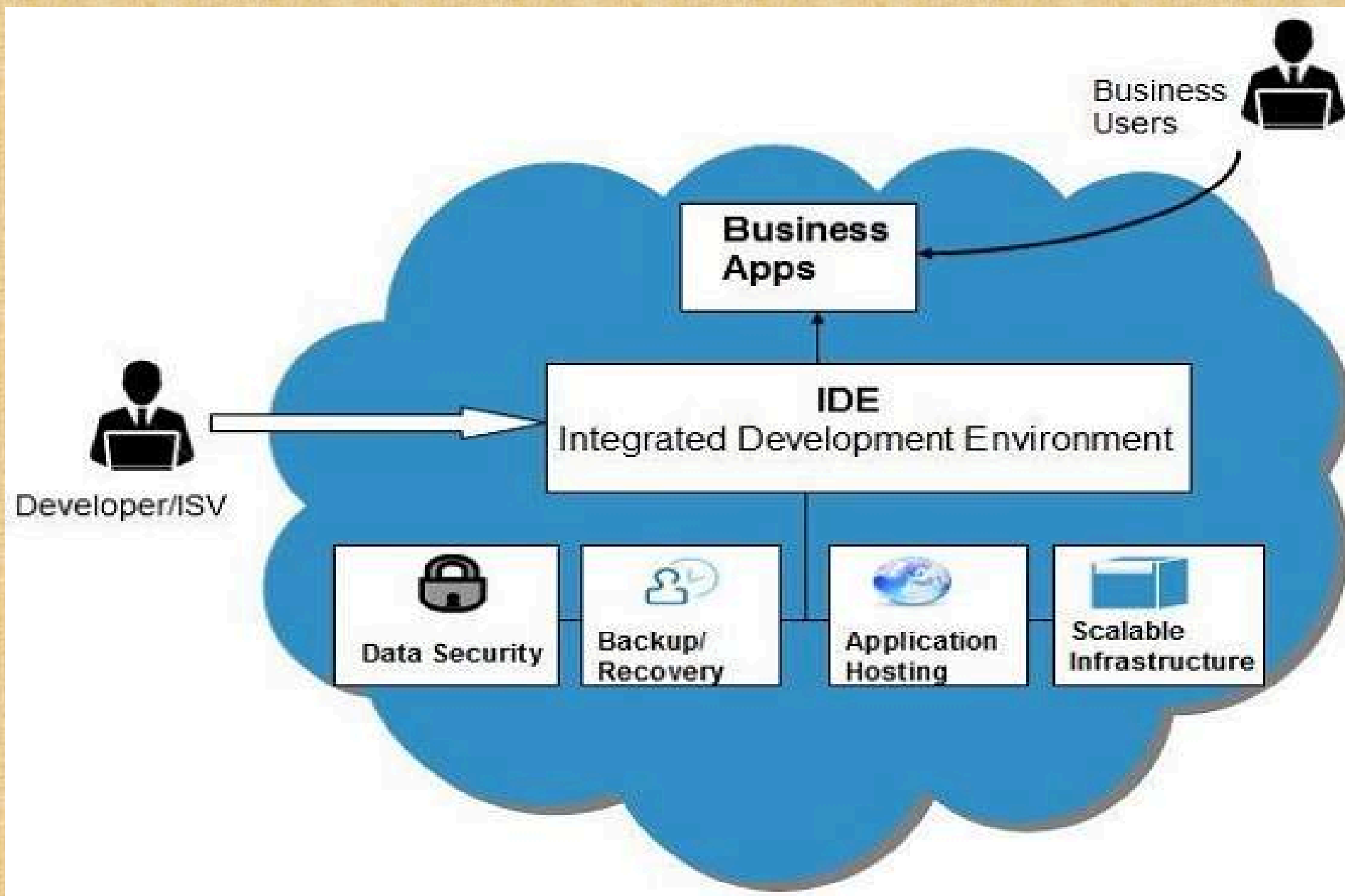
□ Platform as a Service | PaaS

Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications. You can purchase these applications from a cloud service provider on a pay-as-per use basis and access them using the Internet connection. In PaaS, back end scalability is managed by the cloud service provider, so end-users do not need to worry about managing the infrastructure.

PaaS includes infrastructure (servers, storage, and networking) and platform (middleware, development tools, database management systems, business intelligence, and more) to support the web application life cycle.

Example: Google App Engine, Force.com, Joyent, Azure.

- The following diagram shows how PaaS offers an API and development tools to the developers and how it helps the end user to access business applications.





PaaS providers provide the Programming languages, Application frameworks, Databases, and Other tools:

☐ **Programming languages**

PaaS providers provide various programming languages for the developers to develop the applications. Some popular programming languages provided by PaaS providers are Java, PHP, Ruby, Perl, and Go.

☐ **2. Application frameworks**

PaaS providers provide application frameworks to easily understand the application development. Some popular application frameworks provided by PaaS providers are Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.

☐ **3. Databases**

PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.

☐ **4. Other tools**

PaaS providers provide various other tools that are required to develop, test, and deploy the applications.

Advantages of PaaS

There are the following advantages of PaaS –

☐ 1) Simplified Development

PaaS allows developers to focus on development and innovation without worrying about infrastructure management.

☐ 2) Lower risk

No need for up-front investment in hardware and software. Developers only need a PC and an internet connection to start building applications.

☐ 3) Prebuilt business functionality

Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.

☐ 4) Instant community

PaaS vendors frequently provide online communities where the developer can get the ideas to share experiences and seek advice from others.

☐ 5) Scalability

Applications deployed can scale from one to thousands of users without any changes to the applications.

Disadvantages of PaaS:

☐ Vendor lock-in

One has to write the applications according to the platform provided by the PaaS vendor, so the migration of an application to another PaaS vendor would be a problem.

For example, an application written in Python against API of Google, and using App Engine of Google is likely to work only in that environment.

☐ 2) Data Privacy

Corporate data, whether it can be critical or not, will be private, so if it is not located within the walls of the company, there can be a risk in terms of privacy of data.

☐ 3) Integration with the rest of the systems applications

It may happen that some applications are local, and some are in the cloud. So there will be chances of increased complexity when we want to use data which in the cloud with the local data.

□ IaaS (Infra as a Service)

IaaS provides the infrastructure such as virtual machines and other resources like virtual-machine disk image library, block and file-based storage, firewalls, load balancers, IP addresses, virtual local area networks etc. Infrastructure as service or IaaS is the basic layer in cloud computing model.

Common examples: DigitalOcean, Linode, Rackspace, Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE) are some popular examples of IaaS.

□ PaaS (Platform as a Service)

PaaS or platform as a service model provides you computing platforms which typically includes an operating system, programming language execution environment, database, web server. technically It is a layer on top of IaaS as the second thing you demand after Infrastructure is a platform.

Common examples: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos.

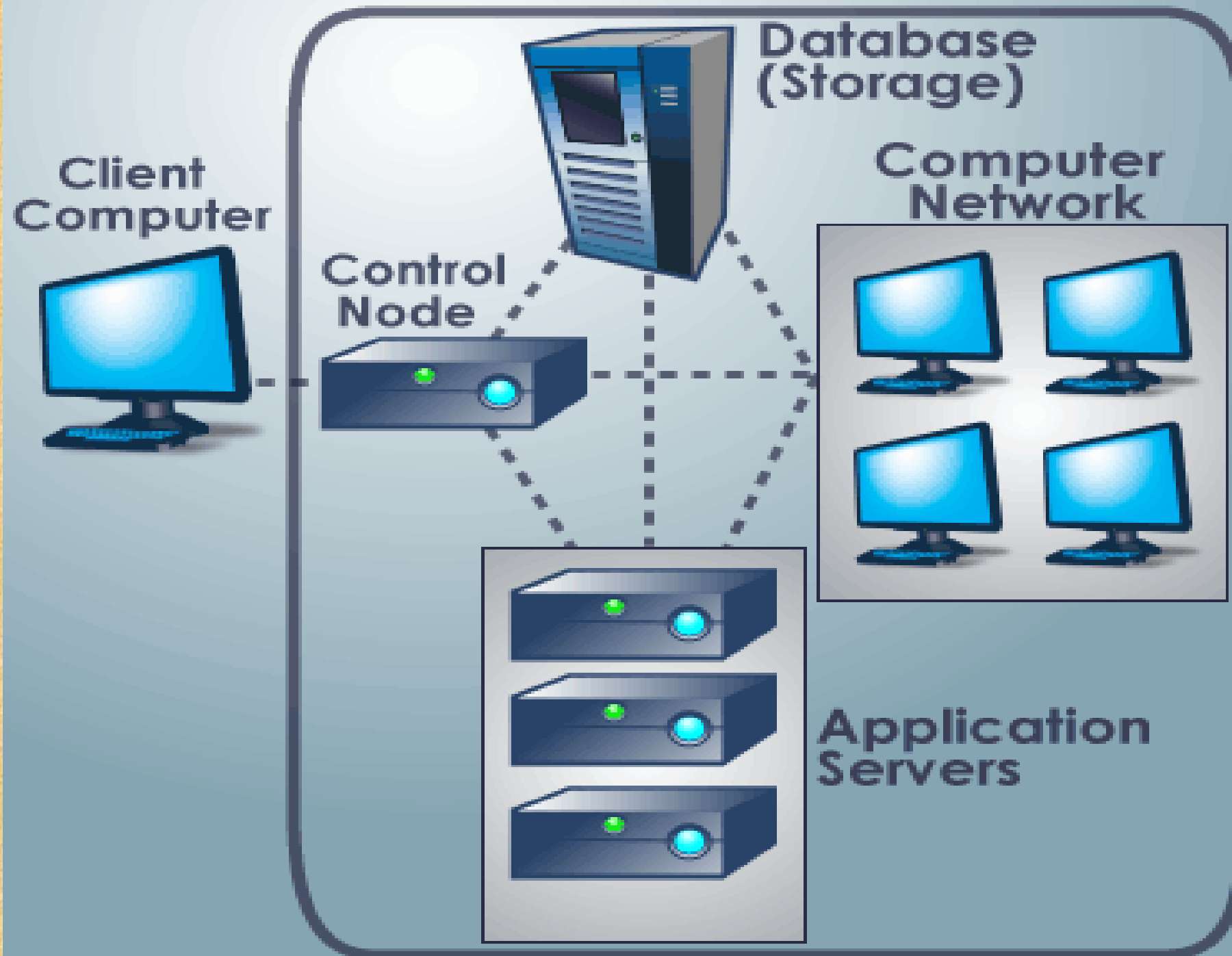
□ SaaS (Software as a Service)

In a SaaS, you are provided access to application services installed at a server. You don't have to worry about installation, maintenance or coding of that software. You can access and operate the software with just your browser. You don't have to download or install any kind of setup or OS, the software is just available for you to access and operate. The software maintenance or setup or help will be provided by SaaS provider company and you will only have to pay for your usage.

Common examples: Google Apps, Microsoft office365, Google docs, Gmail, WHMCS billing software

How Cloud Computing Works

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Deployment Models:

Deployment models define the type of access to the cloud, i.e., how the cloud is located? Or

A **cloud deployment model** is a specific configuration of environment parameters such as the accessibility and proprietorship of the deployment infrastructure and storage size. This means that deployment types vary depending on who controls the infrastructure and where it's located. Cloud can have any of the four types of access:

- ☐ Private cloud
- ☐ Community cloud
- ☐ Public cloud
- ☐ Hybrid cloud

□ Public Cloud

This type of cloud services is provided on a network for public use. Customers have no control over the location of the infrastructure. It is based on a shared cost model for all the users, or in the form of a licensing policy such as pay per user. Public deployment models in the cloud are perfect for organizations with growing and fluctuating demands. It is also popular among businesses of all sizes for their web applications, webmail, and storage of non-sensitive data.

The public cloud deployment model is the first choice for businesses with low privacy concerns. When it comes to popular public cloud deployment models, examples are Amazon Elastic Compute Cloud (Amazon EC2 — the top service provider according to ZDNet), Microsoft Azure, Google App Engine, IBM Cloud, Salesforce Heroku and others.

The Advantages of a Public Cloud

- ☐ **Hassle-free infrastructure management.** Having a third party running your cloud infrastructure is convenient: you do not need to develop and maintain your software because the service provider does it for you. In addition, the infrastructure setup and use are uncomplicated.
- ☐ **High scalability.** You can easily extend the cloud's capacity as your company requirements increase.
- ☐ **Reduced costs.** You pay only for the service you use, so there's no need to invest in hardware or software.
- ☐ **24/7 uptime.** The extensive network of your provider's servers ensures your infrastructure is constantly available and has improved operation time.

The Disadvantages of a Public Cloud

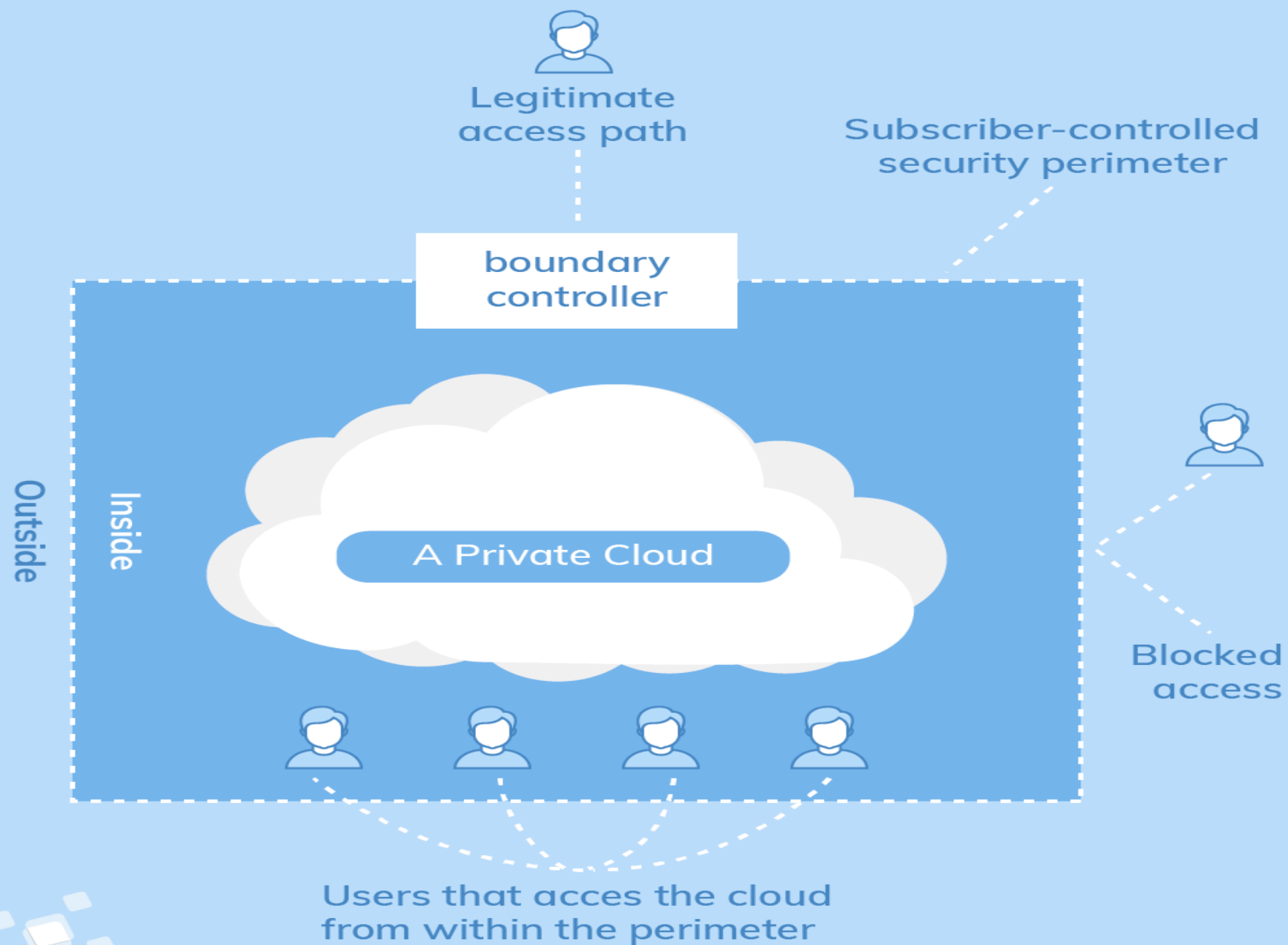
- ❑ **Compromised reliability.** That same server network is also meant to ensure against failure But often enough, public clouds experience outages and malfunction, as in the case of the 2016 [Salesforce CRM disruption](#) that caused a storage collapse.
- ❑ **Data security and privacy issues give rise to concern.** Although access to data is easy, a public deployment model deprives users of knowing where their information is kept and who has access to it.
- ❑ **The lack of a bespoke service.** Service providers have only standardized service options, which is why they often fail to satisfy more complex requirements.

Private Cloud

- There is little to no difference between a public and a private model from the technical point of view, as their architectures are very similar. However, as opposed to a public cloud that is available to the general public, only one specific company owns a private cloud. That is why it is also called an *internal* or *corporate* model.
- The server can be hosted externally or on the premises of the owner company. Regardless of their physical location, these infrastructures are maintained on a designated private network and use software and hardware that are intended for use only by the owner company.
- A clearly defined scope of people have access to the information kept in a private repository, which prevents the general public from using it. In light of numerous breaches in recent years, a growing number of large corporations has decided on a closed private cloud model, as this minimizes data security issues.

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- Compared to the public model, the private cloud provides wider opportunities for customizing the infrastructure to the company's requirements. A private model is especially suitable for companies that seek to safeguard their mission-critical operations or for businesses with constantly changing requirements.
 - Multiple public cloud service providers, including Amazon, IBM, Cisco, Dell and Red Hat, also provide private solutions.

Private Cloud



The Benefits of a Private Cloud:

All the benefits of this deployment model result from its autonomy. They are the following:

- ☐ Bespoke and flexible development and high scalability, which allows companies to customize their infrastructures in accordance with their requirements
- ☐ High security, privacy and reliability, as only authorized persons can access resources

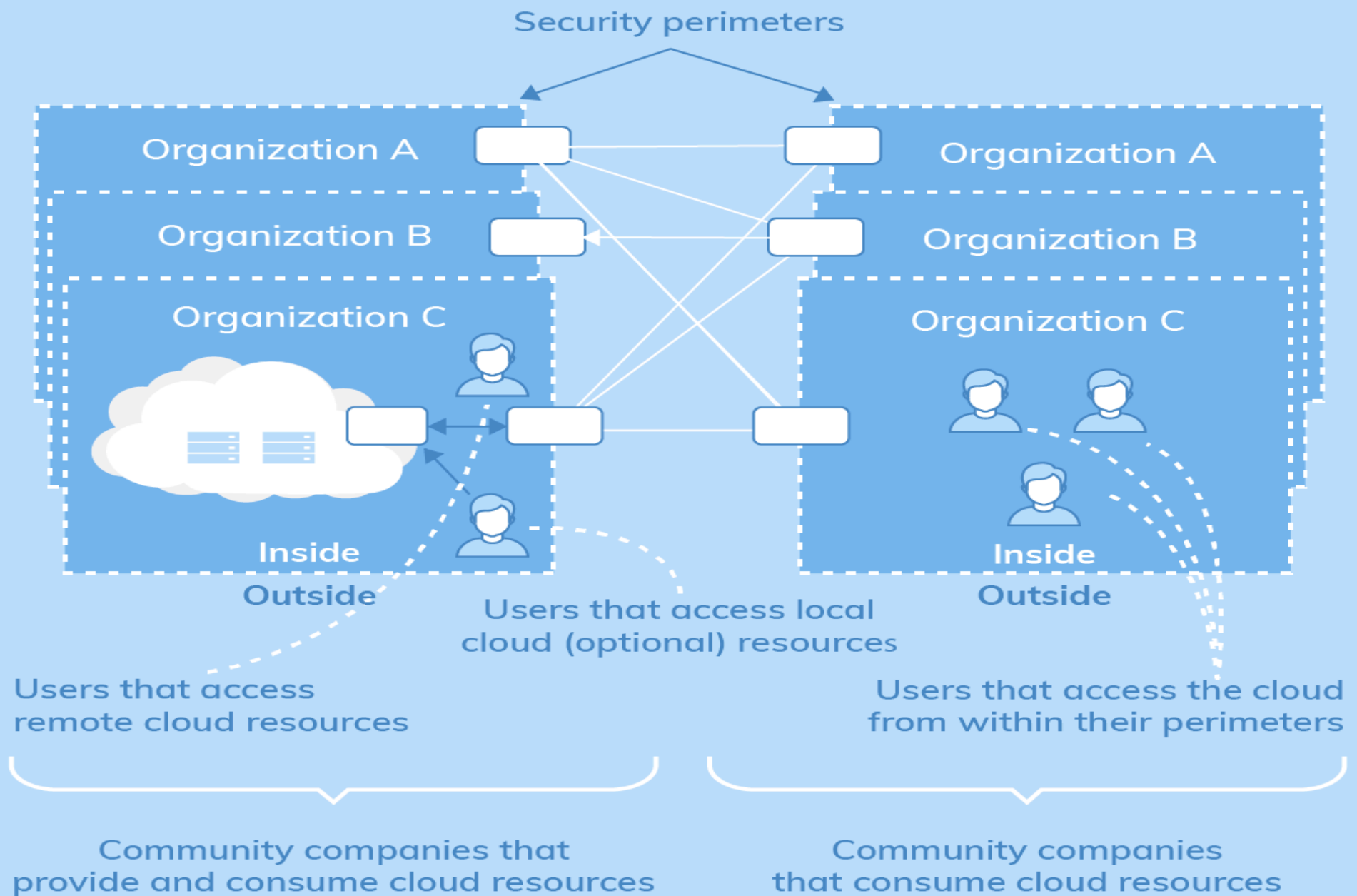
The Drawbacks of a Private Cloud:

- ☐ The major disadvantage of the private cloud deployment model is its cost, as it requires considerable expense on hardware, software and staff training. That is why this secure and flexible computing deployment model is not the right choice for small companies.

Community Cloud

- A community deployment model largely resembles the private one; the only difference is the set of users. Whereas only one company owns the private cloud server, several organizations with similar backgrounds share the infrastructure and related resources of a community cloud.
- If all the participating organizations have uniform security, privacy and performance requirements, this multi-tenant data center architecture helps these companies enhance their efficiency, as in the case of joint projects. A centralized cloud facilitates project development, management and implementation. The costs are shared by all users. Or
- It is a mutually shared model between organizations that belong to a particular community such as banks, government organizations, or commercial enterprises. Community members generally share similar issues of privacy, performance, and security. This type of deployment model of cloud computing is managed and hosted internally or by a third-party vendor.

Community Cloud



The Strengths of a Community Cloud:

- ☐ Cost reduction
- ☐ Improved security, privacy and reliability
- ☐ Ease of data sharing and collaboration

The Shortcomings of a Community Cloud:

- ☐ High cost compared to the public deployment model
- ☐ Sharing of fixed storage and bandwidth capacity

Hybrid Cloud:

- As is usually the case with any hybrid phenomenon, a hybrid cloud encompasses the best features of the abovementioned deployment models (public, private and community). It allows companies to mix and match the facets of the three types that best suit their requirements.
- As an example, a company can balance its load by locating mission-critical workloads on a secure private cloud and deploying less sensitive ones to a public one. The hybrid cloud deployment model not only safeguards and controls strategically important assets but does so in a cost- and resource-effective way. In addition, this approach facilitates data and application portability.

The Benefits of a Hybrid Cloud:

- Improved security and privacy
- Enhanced scalability and flexibility
- Reasonable price

	Public	Private	Community	Hybrid
Ease of setup and use	Easy	Requires IT proficiency	Requires IT proficiency	Requires IT proficiency
Data security and privacy	Low	High	Comparatively high	High
Data control	Little to none	High	Comparatively high	Comparatively high
Reliability	Low	High	Comparatively high	High
Scalability and flexibility	High	High	Fixed capacity	High
Cost-effectiveness	The cheapest	Cost-intensive; the most expensive model	Cost is shared among community members	Cheaper than a private model but more costly than a public one
Demand for in-house hardware	No	Depends	Depends	Depends

Benefits of cloud computing:

☐ Cost

Cloud computing eliminates the capital expense of buying hardware and software and setting up and running on-site datacenters—the racks of servers, the round-the-clock electricity for power and cooling, the IT experts for managing the infrastructure. It adds up fast.

☐ Speed

Most cloud computing services are provided self service and on demand, so even vast amounts of computing resources can be provisioned in minutes, typically with just a few mouse clicks, giving businesses a lot of flexibility and taking the pressure off capacity planning.

☐ Global scale

The benefits of cloud computing services include the ability to scale elastically. In cloud speak, that means delivering the right amount of IT resources—for example, more or less computing power, storage, bandwidth—right when it is needed and from the right geographic location.



□ **Productivity:**

On-site datacenters typically require a lot of “racking and stacking” — hardware setup, software patching, and other time-consuming IT management chores. Cloud computing removes the need for many of these tasks, so IT teams can spend time on achieving more important business goals.

□ **Performance**

The biggest cloud computing services run on a worldwide network of secure datacenters, which are regularly upgraded to the latest generation of fast and efficient computing hardware. This offers several benefits over a single corporate datacenter, including reduced network latency for applications and greater economies of scale.



☐ **Reliability**

Cloud computing makes data backup, disaster recovery and business continuity easier and less expensive because data can be mirrored at multiple redundant sites on the cloud provider's network.

☐ **Security**

Many cloud providers offer a broad set of policies, technologies and controls that strengthen your security posture overall, helping protect your data, apps and infrastructure from potential threats.

Essential Cloud Computing Characteristics:

- On-demand self-service: consumers can unilaterally provision computing capabilities as needed automatically without requiring human interaction with each service provider.
- Broad network access: capabilities are available over the network and accessed through standard mechanism that promote use by heterogeneous thin or thick client platforms.
- Resources pooling: The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand
- Rapid elasticity: capabilities can be elastically provisioned and released to scale rapidly outward and inward commensurate with demand.
- Measured service: cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service.